

A counterexample to Naimark's problem in ZFC

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Abstract

We construct in ZFC a unital infinite-dimensional simple complex C^* -algebra with a faithful tracial state whose nonzero irreducible representations are all unitarily equivalent. This gives a negative answer to Naimark's problem without additional set-theoretic assumptions.

Contents

1	Introduction	2
2	Pure states, excision, and peaking projections	3
2.1	Unique extensions and peaking	4
2.2	Excision and inequivalent states	5
2.3	Expectations from peaking projections	6
3	Reduced HNN adjunctions	7
3.1	The reduced adjunction	8
3.2	Construction from an amalgamated free product	8
3.3	Interchanging adjunctions and taking limits	11
4	Joining two classes while preserving unique extensions	12
5	Closed subiterations and countable determination	16
5.1	States and traces along a chain	16
5.2	The initial algebra and its subiterations	17
5.3	Moving omitted steps to the end	18
5.4	A separable algebra determines each pure state	20
6	The construction in ZFC	20

1 Introduction

For a complex Hilbert space H , let $K(H)$ denote the C^* -algebra of compact operators on H . Every nonzero irreducible representation of $K(H)$ is unitarily equivalent to its defining representation. Naimark's problem asks whether this property characterizes the algebras $K(H)$. We give a negative answer in ZFC.

Theorem 1.1. *There exists a unital infinite-dimensional simple complex C^* -algebra A with a faithful tracial state such that all its nonzero irreducible representations are unitarily equivalent. In particular, A is not isomorphic to $K(H)$ for any complex Hilbert space H .*

Here a representation is irreducible if it is nonzero and has no nontrivial closed invariant subspace. A tracial state τ is faithful if $\tau(a^*a) > 0$ whenever $a \neq 0$. The last assertion of the theorem follows from the first: $K(H)$ is unital only when H is finite-dimensional. All algebras and Hilbert spaces in the proof are complex, and no separability assumption is placed on A .

History and scope

Naimark's question dates to 1951; see the historical account of the question and Rosenberg's affirmative work in [2]. When the unique nonzero irreducible representation acts on a separable Hilbert space, the algebra is isomorphic to the compact operators; see [5, Proposition 7.2]. This condition concerns an irreducible representation, not merely a faithful representation on a separable Hilbert space. Akemann and Weaver constructed a counterexample assuming Jensen's diamond principle [2, Theorem 5]. They also established independence for the existence of a counterexample generated by $\aleph_1$ elements [2, Corollary 7]. The unrestricted existence question does not impose this bound on the number of generators.

Subsequent work refined the set-theoretic hypotheses and the possible structure of counterexamples. Calderón and Farah obtain a counterexample from $\diamond_{\text{Cohen}}$ together with CH [5, Theorem 5.4]. Under diamond, Vaccaro constructs counterexamples whose tracial state spaces are affinely homeomorphic to any prescribed metrizable Choquet simplex [13, 14]. Thus the presence of a trace is already compatible with the earlier constructions.

Tanaka gives another ZFC construction with a faithful tracial state [10, Theorems 1.1, 4.1 and 5.3], so the existence conclusion of Theorem 1.1 is already present there. His construction starts from the CAR algebra and uses strong shell sums and rank-one defects. It also gives a faithful representation on a separable Hilbert space; that representation is not irreducible. Here we give an alternative proof organized around a selective bridge for faithfully tracial bases without finite-dimensional irreducible representations, reduced-HNN reordering, and dependency-closed unique extension and countable determination. The precise hypotheses and conclusions of these intermediate results are stated below. Our construction has length $\mathfrak{c}^+$, where $\mathfrak{c} = 2^{\aleph_0}$; we make no assertion that this length is the density of the resulting algebra.

The proof in outline

We work with pure states, whose Gelfand–Naimark–Segal (GNS) representations are irreducible. Two pure states are called equivalent when their GNS representations are unitarily equivalent. The construction repeatedly joins a chosen pair of equivalence classes. For this procedure to be controllable, each old pure state must have a unique state extension, and no other pair of old classes may become equivalent.

The main analytic step starts with a faithfully tracial algebra having no finite-dimensional irreducible representations. It adjoins a unitary carrying one decreasing projection sequence to another. Each sequence specifies its state as the only state taking value one on every projection; we call it a *peaking sequence*. When the two states are inequivalent and the sequences have matching trace values strictly decreasing to zero, the resulting reduced HNN extension has exactly the required pure-state properties (Theorem 4.1). The extension is built from a reduced amalgamated free product using Ueda’s matrix-corner construction [11, 12]. Pure-state excision [3] and the cross-excision estimate of Akemann and Weaver [2, Lemma 1] turn uniqueness of extension and the exclusion of unintended joins into norm-decay estimates for reduced words. The estimates apply to every old pure state.

The second step makes every pure state accessible to the recursion. We start with a separable simple algebra, adjoin one unitary at each active successor stage, and take norm closures of unions at limits. Each adjunction depends on only countably many earlier generators. If a set of generators contains all these dependencies, we prove that every pure state on the algebra it generates has a unique extension to the whole algebra (Theorem 5.5). To prove this, we postpone an omitted adjunction past the retained ones. The no-other-join assertion of the bridge keeps its two states inequivalent until the postponed adjunction is performed. For every pure state of the full algebra, excision provides a separable subalgebra generated by a countable dependency-closed set of steps on which the restriction is pure. The closed-subiteration theorem then makes the original state its unique extension to the full algebra (Lemma 5.6).

This last conclusion replaces the state-guessing role of a set-theoretic prediction principle. At every stage below $\mathfrak{c}^+$ there are at most $\mathfrak{c}$ pure states, so a fixed schedule can eventually treat them all. On each separable simple determining subalgebra, the homogeneity theorem of Kishimoto, Ozawa and Sakai [8] supplies the next peaking sequence. A final countable-determination argument catches every pure state of the limit algebra.

The ingredients have several useful precedents. Akemann and Weaver combine pure-state homogeneity with extensions joining prescribed states [2, Corollary 3]. Farah and Hirshberg realize prescribed countable equivalence patterns with unique pure-state extensions [6, Lemma 2.3]; Vaccaro’s corrected construction combines this method with control of the trace space [13, 14]. Akemann, Wassermann and Weaver prove pure-state restriction and unique-extension results for reduced free group algebras [1, Proposition 2.3 and Theorem 2.6]. Calderón and Farah propose using separable subalgebras that uniquely determine the relevant states, followed by an amalgamation preserving unique extension [5, Section 8, especially Question 8.1]. The two features established here are the selective bridge for all old pure states and its compatibility with dependency-closed subiterations. Together they give the countable determination needed for the unrestricted recursion.

Section 2 proves the state-theoretic facts used below. Section 3 constructs the reduced adjunction, and Section 4 proves its pure-state properties. Section 5 establishes countable determination; Section 6 carries out the final recursion.

2 Pure states, excision, and peaking projections

The construction must preserve unique state extensions while changing selected equivalence classes. This section records the state-theoretic tools for doing so. Its main analytic fact is excision: near the support of a pure state, compression of an element approaches its scalar state value.

All algebras and Hilbert spaces are complex. Unless stated otherwise, algebras are unital and inclusions preserve the unit. A *state* on an algebra B is a positive linear functional of norm

one, and it is *pure* if it is an extreme point of the state space. Write (π_h, H_h, ξ_h) for the GNS representation of a state h . Thus ξ_h is a cyclic unit vector and $h(a) = \langle \pi_h(a)\xi_h, \xi_h \rangle$; purity is equivalent to irreducibility of π_h . For pure states, write $h \sim k$ when π_h and π_k are unitarily equivalent. We use $\text{Ad}(u)(a) = uau^*$ for conjugation by a unitary.

A *tracial algebra* is a unital C^* -algebra equipped with a faithful tracial state τ : one has $\tau(ab) = \tau(ba)$, and $\tau(a^*a) = 0$ implies $a = 0$. Traces and other bounded functionals are extended normally to the enveloping von Neumann algebra B^{**} , with the same notation. The normal extension of a trace is tracial, by separate ultraweak continuity. For a representation π of B , its normal extension to B^{**} is denoted by π^{**} .

For an inclusion $B \subset C$, we shall use the property

$$\text{every pure state of } B \text{ has exactly one state extension to } C. \quad (\text{U})$$

Here uniqueness is among all states. States always extend across a unital inclusion, by the positive form of the Hahn–Banach theorem.

2.1 Unique extensions and peaking

Lemma 2.1. *If a pure state h of B has a unique state extension $\tilde{h}$ to $C \supset B$, then $\tilde{h}$ is pure.*

Proof. In a nontrivial convex decomposition of $\tilde{h}$, both restrictions to B equal h by purity. Uniqueness then makes both states equal to $\tilde{h}$. $\square$

Lemma 2.2. *Equivalent pure states of a unital C^* -algebra B are conjugate by a unitary of B . Consequently, if they each have a unique state extension to an algebra $C \supset B$, their extensions are equivalent.*

Proof. Realize the two states by unit vectors ξ, η in a common irreducible representation π . On their finite-dimensional span choose a unitary carrying ξ to η and a self-adjoint logarithm of that unitary. Kadison transitivity, including its self-adjoint form [7, Theorem 1], supplies a self-adjoint $a \in B$ whose action on this span is the chosen logarithm. The span reduces $\pi(a)$, so $u = e^{ia}$ satisfies $\pi(u)\xi = \eta$. If h is represented by ξ and k by η , then $k = h \circ \text{Ad}(u^*)$. Conjugating their unique extensions by the same unitary proves the last assertion. $\square$

A decreasing sequence of projections $(P_n)_{n \geq 0}$ in B *peaks to* a state h if h is the only state satisfying $h(P_n) = 1$ for every n . Such a state is pure: both states in any convex decomposition satisfy the same constraints. The following elementary observation will let us pass to smaller algebras without assuming (U) for the inclusion.

Lemma 2.3 (Peaking under restriction). *Let $B \subset C$ be a unital inclusion and let (P_n) be a decreasing sequence of projections in B that peaks to a state H of C . Then (P_n) peaks to $h = H|_B$ in B , and H is the unique state extension of h to C .*

Proof. Every state of B satisfying the projection constraints extends to a state of C with the same constraints, hence to H . It must therefore equal h . Every extension of h also satisfies the constraints and must equal H . $\square$

2.2 Excision and inequivalent states

For a state h , its *support projection* $r_h \in B^{**}$ is the least projection on which its normal extension takes value one. We shall prove that this projection is minimal when h is pure, together with the norm approximation that we need. The decreasing-net form of pure-state excision used here is proved by Akemann, Anderson, and Pedersen [3, Proposition 2.2]. We record a proof in the net form used below.

Lemma 2.4 (Decreasing excision). *Let h be a pure state of a unital C^* -algebra B . There is a decreasing net of positive contractions $x_\lambda \in B$ such that $h(x_\lambda) = 1$, x_λ decreases strongly to the minimal projection r_h in B^{**} , and, for every $a \in B$,*

$$\|x_\lambda(a - h(a)1)x_\lambda\| \longrightarrow 0. \quad (2.1)$$

Moreover, $r_h a r_h = h(a)r_h$ for every $a \in B$.

Proof. Put

$$L = \{a \in B : h(a^*a) = 0\}, \quad N = L \cap L^*.$$

Here L is a closed left ideal and N is a C^* -subalgebra. If $a \in L$, then $a^*a \in N$, since $h((a^*a)^2) \leq \|a\|^2 h(a^*a) = 0$.

First, h is the only state annihilating N . If another state k annihilates N , then L is contained in the kernel of $a \mapsto \pi_k(a)\xi_k$. The map $Q_h : B \rightarrow H_h$, $Q_h(a) = \pi_h(a)\xi_h$, is onto by one-vector Kadison transitivity. This assertion concerns the whole Hilbert space, even when it is nonseparable. Since $\ker Q_h = L$, the open mapping theorem for $B/L \rightarrow H_h$ shows that

$$T(\pi_h(a)\xi_h) = \pi_k(a)\xi_k$$

defines a bounded operator $T : H_h \rightarrow H_k$. It intertwines the representations. Thus T^*T is scalar by irreducibility of π_h ; its value on ξ_h shows that $T^*T = 1$. Taking vector coefficients gives $k = h$.

We next construct the decreasing net. Order N_+ by the usual operator order, a directed order, and set

$$e_s = 1 - (1 + s)^{-1}, \quad x_s = 1 - e_s \quad (s \in N_+).$$

Functional calculus gives $e_s \in N$, and operator monotonicity of inversion makes (e_s) increasing. For $b \in N$ and $s \geq m(bb^* + b^*b)$,

$$\|(1 - e_s)b\|^2 \leq \frac{1}{m} \|(1 + s)^{-1}s(1 + s)^{-1}\| \leq \frac{1}{4m};$$

the same estimate holds for $\|b(1 - e_s)\|^2$. Thus (e_s) is an approximate identity of N , and $h(x_s) = 1$.

Let q be the strong limit of (e_s) , the support projection of N . Every normal state supported on $1 - q$ annihilates N , and hence is h ; conversely, $h(1 - q) = 1$. The corner $(1 - q)B^{**}(1 - q)$ therefore has exactly one normal state. Normal states separate its self-adjoint elements, so this corner is scalar. It follows that $1 - q = r_h$ is minimal and $x_s \downarrow r_h$. The identity $r_h a r_h = h(a)r_h$ follows by applying h to this scalar corner.

Finally, let ρ_s be arbitrary states of B . Every weak-star cluster point of the positive functionals $a \mapsto \rho_s(x_s a x_s)$ annihilates N , by the approximate-identity estimates. A nonzero cluster point, after normalization, is therefore h . For self-adjoint a , a failure of (2.1) would give a subnet and norming states ρ_s with a cluster point nonzero on $a - h(a)1$, a contradiction. Real and imaginary parts give the conclusion for arbitrary a . $\square$

Lemma 2.5 (Projection excision). *If decreasing projections (P_n) in B peak to h , then $P_n \downarrow r_h$ strongly and*

$$\|P_n(a - h(a)1)P_n\| \longrightarrow 0 \quad (a \in B).$$

In particular, if $h(a) = 0$, then $\|P_n a P_t\| \rightarrow 0$ on the product index set.

Proof. The infimum of the P_n in B^{**} supports exactly one normal state, namely h . As in the preceding proof, it is the minimal projection r_h . For arbitrary states ρ_n , every cluster point ψ of $\rho_n(P_n \cdot P_n)$ vanishes on $1 - P_m$ for each fixed m . If $\psi \neq 0$, its normalization satisfies every peaking constraint and therefore equals h . The same norming-state argument proves excision. For the last assertion use nesting:

$$\|P_n a P_t\| \leq \|P_{\min(n,t)} a P_{\min(n,t)}\|. \quad \square$$

Excision also separates inequivalent states in norm. We use the following form of the cross-excision estimate of Akemann and Weaver [2, Lemma 1].

Lemma 2.6 (Cross-excision). *Let $h \not\sim k$ be pure states of B , and let (x_λ) and (y_ν) be decreasing nets as in Lemma 2.4, for h and k respectively. Then, for every $a \in B$,*

$$\|x_\lambda a y_\nu\| \longrightarrow 0 \quad (2.2)$$

on the product directed set. Either net may be a peaking sequence of projections.

Proof. The normal extension of π_h annihilates r_k . Otherwise a unit vector in the range of $\pi_h^{**}(r_k)$ would represent k , by $r_k a r_k = k(a)r_k$. Every nonzero vector in an irreducible representation is cyclic, so this would imply $h \sim k$.

Consequently $h(a y_{\nu_0} a^*) \rightarrow 0$ as ν_0 increases. For $\nu \geq \nu_0$, the inequalities $y_\nu^2 \leq y_\nu \leq y_{\nu_0}$ give

$$\begin{aligned} \|x_\lambda a y_\nu\|^2 &\leq \|x_\lambda a y_{\nu_0} a^* x_\lambda\| \\ &\leq h(a y_{\nu_0} a^*) + \|x_\lambda (a y_{\nu_0} a^* - h(a y_{\nu_0} a^*) 1) x_\lambda\|. \end{aligned}$$

Choose ν_0 first, then use excision to choose λ_0 . This proves smallness on a rectangular tail and hence the product-net limit. Lemma 2.5 gives the stated projection case. $\square$

Lemma 2.7 (Trace-null supports). *If B has no finite-dimensional irreducible representations and τ is a tracial state on B , then $\tau(r_h) = 0$ for every pure state h .*

Proof. The space H_h is infinite-dimensional. Choose orthonormal unit vectors η_j and unitaries $u_j \in B$ with $\pi_h(u_j)\xi_h = \eta_j$, as in Lemma 2.2. The projections $u_j r_h u_j^*$ are pairwise orthogonal, since

$$r_h u_i^* u_j r_h = h(u_i^* u_j) r_h = 0 \quad (i \neq j).$$

They all have trace $\tau(r_h)$, so $m\tau(r_h) \leq 1$ for every positive integer m . $\square$

2.3 Expectations from peaking projections

We now use projection excision to construct the conditional expectations needed for the tracial extensions. A *conditional expectation* $E : C \rightarrow B$ is a unital completely positive map fixing B and satisfying $E(b_1 a b_2) = b_1 E(a) b_2$ for $b_1, b_2 \in B$. It is *faithful* if $E(a^* a) = 0$ implies $a = 0$.

Lemma 2.8. *For a unital subalgebra B of a faithfully tracial algebra (C, τ) , there is at most one trace-preserving conditional expectation $C \rightarrow B$. If one exists, it is faithful.*

Proof. For such an expectation, bimodularity gives $\tau(b^*E(a)) = \tau(b^*a)$ for every $b \in B$. Two expectations therefore have difference $d \in B$ satisfying $\tau(d^*d) = 0$, so they agree. Faithfulness follows from $\tau(E(a^*a)) = \tau(a^*a)$. $\square$

Lemma 2.9. *Let (B, τ) be faithfully tracial. Suppose decreasing projections $(p_n)_{n \geq 0}$ peak to h , with $p_0 = 1$ and trace values strictly decreasing to zero. Then $D = C^*(p_n : n \geq 0)$ is isomorphic to $C(\mathbb{N}_0 \cup \{\infty\})$, and there is a trace-preserving conditional expectation $E : B \rightarrow D$. Writing $d_n = p_n - p_{n+1}$ for the finite-point atoms, it is given by*

$$E(a)(n) = \frac{\tau(d_n a d_n)}{\tau(d_n)}, \quad E(a)(\infty) = h(a). \quad (2.3)$$

Proof. The d_n are nonzero orthogonal projections. Their closed linear span is isomorphic to $c_0(\mathbb{N}_0)$, and adjoining the unit gives the claimed description of D . Indeed $p_n = 1 - \sum_{j < n} d_j$ and $\|p_n\| = 1$, since $\tau(p_n) > 0$. Multiplying by p_n shows that 1 has distance at least one from every linear combination of $d_0, \dots, d_{n-1}$.

The values in (2.3) define a continuous function: by $d_n \leq p_n$ and Lemma 2.5,

$$\left| \frac{\tau(d_n a d_n)}{\tau(d_n)} - h(a) \right| \leq \|p_n(a - h(a)1)p_n\| \rightarrow 0.$$

Every coordinate of E is a state, so E is unital and completely positive: positivity of each matrix amplification is tested pointwise in the commutative range. The map fixes D . To check bimodularity, an element of D acts scalarly on each d_n ; at infinity, $h|_D$ is the evaluation character. Cauchy-Schwarz for h then gives $h(ba) = h(b)h(a)$ and $h(ab) = h(a)h(b)$ for $b \in D$. These coordinate identities prove bimodularity.

Finally, $\sum_n \tau(d_n) = 1$ and the trace on D has no mass at infinity. Summing the finite coordinates gives $\tau(E(a)) = \tau(a)$, since the remaining trace satisfies $|\tau(p_N a)| \leq \|a\| \tau(p_N) \rightarrow 0$. $\square$

Unique state extensions and a faithful expectation will also preserve simplicity, without a separate analysis of ideals in each extension.

Lemma 2.10. *Suppose $B \subset C$ satisfies (U), B is simple, and there is a faithful conditional expectation $E : C \rightarrow B$. Then C is simple.*

Proof. Let J be a proper closed two-sided ideal of C . The quotient map $q : C \rightarrow C/J$ is injective on B , since B is simple and $q(1) \neq 0$. Every pure state h of B therefore extends through this quotient to a state of C annihilating J . By (U), this extension is $h \circ E$. Thus $h(E(a)) = 0$ for every positive $a \in J$ and every pure h on B . Pure states separate positive elements, so $E(a) = 0$; faithfulness gives $a = 0$. Hence $J = 0$. $\square$

3 Reduced HNN adjunctions

We now construct the extension that will join two pure-state classes. The construction itself has a more general input: two trace-preserving isomorphic subalgebras of a faithfully tracial algebra. It adjoins a unitary implementing the isomorphism, while prescribing the expectation of every word in that unitary. These prescriptions will also let us restrict the construction to smaller bases and interchange two adjunctions with data in the same base.

The matrix-corner construction below is the reduced HNN construction of Ueda; see [11, Proposition 2.2 and Remark 2.3] and [12]. We give the details, including the faithful representations and expectations needed later. The corresponding expectation and embedding results for reduced amalgamated free products appear in [4, Lemma 1.1 and Theorem 1.3].

3.1 The reduced adjunction

Let (A, τ) be a unital C^* -algebra with faithful tracial state. Fix unital subalgebras $D_1, D_2 \subset A$, trace-preserving conditional expectations $E_i : A \rightarrow D_i$, and a trace-preserving isomorphism $\theta : D_1 \rightarrow D_2$. A word in A and a unitary v has the form

$$a_0 v^{\epsilon_1} a_1 \cdots v^{\epsilon_m} a_m, \quad a_t \in A, \quad \epsilon_t \in \{+1, -1\}. \quad (3.1)$$

For $m \geq 1$, call it *reduced* if every internal coefficient satisfies

$$\begin{cases} E_1(a_t) = 0, & (\epsilon_t, \epsilon_{t+1}) = (+1, -1), \\ E_2(a_t) = 0, & (\epsilon_t, \epsilon_{t+1}) = (-1, +1). \end{cases} \quad (3.2)$$

There is no condition at equal consecutive signs.

Proposition 3.1. *Let (A, τ) be a unital C^* -algebra with faithful tracial state, let $D_1, D_2 \subset A$ be unital subalgebras with trace-preserving conditional expectations $E_i : A \rightarrow D_i$, and let $\theta : D_1 \rightarrow D_2$ be a trace-preserving isomorphism. There is a unital extension with a faithful tracial state extending τ ,*

$$H(A) = C^*(A, v), \quad vdv^* = \theta(d) \quad (d \in D_1),$$

where v is called the stable unitary, and a trace-preserving conditional expectation $E_A : H(A) \rightarrow A$ annihilating every nonempty reduced word (3.1). Among faithfully tracial extensions generated by A and such a unitary, the relation and vanishing of the trace on nonempty reduced words determine $H(A)$ uniquely, by an isomorphism fixing A and v .

We construct this adjunction as a matrix corner of an amalgamated free product. The proof below records the module and its two compression expectations; these will also be used in the pure-state bridge of Section 4.

3.2 Construction from an amalgamated free product

An element is *centered* for a conditional expectation E if it belongs to $\ker E$.

Lemma 3.2. *Let F and G be unital C^* -algebras with faithful tracial states, containing a common unital subalgebra D on which the traces agree. Suppose that $E_D^F : F \rightarrow D$ and $E_D^G : G \rightarrow D$ are trace-preserving conditional expectations. There is a unital C^* -algebra Z generated by faithful copies of F and G agreeing on D , with a faithful tracial state extending both traces and faithful trace-preserving conditional expectations*

$$E_D : Z \longrightarrow D, \quad E_F : Z \longrightarrow F, \quad E_D = E_D^F \circ E_F.$$

The map E_D vanishes on every nonempty alternating product of elements of $\ker E_D^F$ and $\ker E_D^G$. The map E_F vanishes on every product

$$a_0 b_1 a_1 \cdots b_m a_m, \quad m \geq 1, \quad (3.3)$$

where $a_0, a_m \in F$, $b_t \in \ker E_D^G$, and $a_t \in \ker E_D^F$ for $0 < t < m$. The linear span of F and these products is dense in Z .

Proof. For $i \in \{F, G\}$, complete i as a right Hilbert D -module L_i with inner product

$$\langle x, y \rangle_D = E_D^i(x^*y).$$

Left multiplication by $a \in i$ is bounded, since

$$E_D^i(x^*a^*ax) \leq \|a\|^2 E_D^i(x^*x),$$

and its adjoint is left multiplication by a^* . The expectation gives an orthogonal decomposition

$$L_i = D \oplus L_i^\circ, \quad L_i^\circ = \overline{\ker E_D^i}.$$

Both summands are invariant under left and right multiplication by D . Form the Hilbert D -module

$$K = D \oplus \bigoplus_{\substack{m \geq 1, \\ i_t \in \{F, G\} \\ i_t \neq i_{t+1}}} L_{i_1}^\circ \otimes_D \cdots \otimes_D L_{i_m}^\circ. \quad (3.4)$$

Here $\otimes_D$ is the interior tensor product of Hilbert modules; the left D -action on a word acts on its first factor. The first summand D is called the vacuum summand, with distinguished vector 1.

Let $K^{(i)}$ be the vacuum summand together with all words not starting with i . The decomposition of L_i identifies

$$K = L_i \otimes_D K^{(i)}.$$

Represent i by left multiplication on the first tensor factor. The two representations agree on D . They are faithful: an element a annihilating the vacuum satisfies $E_D^i(a^*a) = 0$, so its faithful factor trace gives $a = 0$. Let Z be the C^* -algebra they generate in the adjointable operators on K .

A centered element of i acting on $K^{(i)}$ adds its own tensor factor at the front. Thus a nonempty alternating centered product has zero vacuum compression. Centering arbitrary factor elements shows that D and these products span a dense subspace of Z . Vacuum compression consequently takes values in the left copy of D and defines the conditional expectation E_D .

Write τ_D for the common trace on D . We next prove that $\tau_Z = \tau_D \circ E_D$ is tracial and faithful. For traciality, compare $\tau_Z(aw)$ and $\tau_Z(wa)$ when a belongs to one factor and w is an alternating centered word. If w has length at least two, both values vanish. Indeed, multiplication at a same-factor end, followed by centering that end product, leaves either an alternating centered word or a D -coefficient times a nonempty such word. At a different-factor end, first center a and use D -bimodularity. For a one-letter word, use the factor trace when the factors agree; when they differ, both values are zero. For $w \in D$, use the factor trace. Density and repeated cyclic movement of a factor now prove traciality on all of Z .

Suppose $\tau_Z(z^*z) = 0$. For every $t \in Z$, traciality gives

$$0 \leq \tau_Z(t^*z^*zt) = \tau_Z(ztt^*z^*) \leq \|t\|^2 \tau_Z(zz^*) = 0.$$

The trace on D is faithful, so $E_D((zt)^*(zt)) = 0$. This is the D -valued inner product of $zt1$ with itself, and hence $zt1 = 0$. Products of centered letters create all elementary tensors in (3.4); thus $Z1$ is dense in K . It follows that $z = 0$, proving faithfulness of τ_Z .

Finally, let P_F be the orthogonal projection onto $D \oplus L_F^\circ = L_F$. This submodule reduces the F -action. A core $w = b_1a_1 \cdots a_{m-1}b_m$ as in (3.3) sends L_F into words starting with G , so $P_FwP_F = 0$. Since P_F commutes with F , arbitrary endpoint coefficients satisfy

$$P_Fa_0wa_mP_F = a_0P_FwP_Fa_m = 0.$$

Successive centering shows that F and the products (3.3) span a dense subspace. Their compressions belong to the faithful left copy of F on L_F , which is norm closed. Compression therefore defines a conditional expectation $E_F : Z \rightarrow F$. The vacuum lies in L_F , giving $E_D = E_D^F \circ E_F$ and trace preservation. Faithfulness follows from faithfulness of τ_Z . $\square$

Proof of Proposition 3.1. Take

$$F = M_2(A), \quad G = M_2(D_1), \quad D = D_1 \oplus D_2.$$

Embed D in F by $(d_1, d_2) \mapsto \text{diag}(d_1, d_2)$ and in G by $(d_1, d_2) \mapsto \text{diag}(d_1, \theta^{-1}(d_2))$. The normalized matrix traces agree on D , and the maps

$$E_D^F(a) = (E_1(a_{11}), E_2(a_{22})), \quad E_D^G(b) = (b_{11}, \theta(b_{22}))$$

are trace-preserving conditional expectations. Apply Lemma 3.2 to obtain Z , and retain the module notation used in its proof. The diagonal matrix projections $e_i = e_{ii}$ are common to the two factors. Set

$$s = e_{21}^G, \quad H(A) = e_1 Z e_1, \quad v = e_{12}^F s, \quad (3.5)$$

and identify A with $e_1 F e_1$. Then $v^* v = v v^* = e_1$. For $d \in D_1$, the element $s d s^*$ is the second-corner copy of $\theta(d)$ in F , so $v d v^* = \theta(d)$ in the first corner.

The matrix units of F identify Z with $M_2(e_1 Z e_1)$. Under this identification, the matrix algebra over $C^*(A, v)$ contains F and $s = e_{21}^F v$. It contains G as well, because G is generated by D and s, s^* . Hence $H(A) = C^*(A, v)$. The trace on the corner is $2\tau_Z$, and its restriction to A is τ . The restriction of E_F gives the required faithful trace-preserving expectation onto A .

To check reduced words, write $q = e_{12}^F$, so $v = q s$ and $v^* = s^* q^*$. Between consecutive G -letters the internal F -coefficients are

$(\epsilon_t, \epsilon_{t+1})$	$(+, +)$	$(+, -)$	$(-, +)$	$(-, -)$
coefficient	$a_t q$	a_t	$q^* a_t q$	$q^* a_t$

The first and last are off diagonal. The middle two have zero D -expectation precisely under the respective conditions in (3.2). Since s, s^* are centered, the expanded word is a product of the form (3.3), and its expectation onto F , hence onto A , is zero.

For uniqueness, split a coefficient at an opposite-sign turn as $(a - E_i(a)) + E_i(a)$. The first term is centered at that turn, while the second cancels two stable letters using the unitary relation. Repeating this operation expresses every polynomial as an A -term plus nonempty reduced words. Thus the stipulated data determine the trace of every polynomial, and in particular all moments of $x^* x$ for each polynomial x .

For a faithful tracial state t on any C^* -algebra,

$$\|x\| = \lim_{r \rightarrow \infty} t((x^* x)^r)^{1/(2r)}. \quad (3.6)$$

To see this, restrict t to the commutative algebra generated by $x^* x$ and the unit. Its spectral measure has full support, by faithfulness, and its r th-root moments tend to the spectral maximum. Consequently identical polynomial trace moments give identical polynomial norms. The generator correspondence is therefore isometric and extends to the asserted isomorphism. This argument requires no expectation in the comparison extension beyond the stated trace-vanishing rule. $\square$

We will also use the explicit spanning description of the matrix algebra Z . In the model (3.5), every centered G -element is a sum of D -multiples of s and s^* . Absorb these D -coefficients into neighboring F -coefficients; their internal centering is preserved by bimodularity. Thus F and the products

$$a_0 b_1 a_1 \cdots b_m a_m, \quad m \geq 1, \quad b_t \in \{s, s^*\}, \quad a_t \in F, \quad E_D^F(a_t) = 0 \quad (0 < t < m) \quad (3.7)$$

have dense linear span in Z .

Corollary 3.3. *With the hypotheses of Proposition 3.1, let $A' \subset A$ be a unital subalgebra containing D_1 and D_2 . Then $C^*(A', v) \subset H(A)$ is the reduced adjunction $H(A')$ formed with the restricted trace and expectations $E_i|_{A'}$. The expectation E_A restricts to the expectation $H(A') \rightarrow A'$.*

Proof. Because $D_i \subset A'$, the restrictions $E_i|_{A'}$ remain conditional expectations onto D_i . The subalgebra $C^*(A', v)$ inherits a faithful trace, the unitary relation and the reduced-word trace vanishing. Proposition 3.1 identifies it with $H(A')$. The reduction of polynomials in the preceding proof shows that E_A maps this subalgebra into A' and has the stated restriction. $\square$

3.3 Interchanging adjunctions and taking limits

The construction now has its two required analytic features: a faithful expectation onto the base and a complete prescription of reduced-word moments. The latter makes it possible to change the order of two adjunctions whose data were already present in a common base.

Proposition 3.4. *Let O be a unital faithfully tracial C^* -algebra. For each $t \in \{j, k\}$, let $D_1^t, D_2^t \subset O$, $E_i^t : O \rightarrow D_i^t$ and $\theta_t : D_1^t \rightarrow D_2^t$ satisfy the hypotheses of Proposition 3.1. When forming an adjunction over $H_j(O)$ or $H_k(O)$, extend the other data expectations by composition with the expectation onto O . There is a trace-preserving isomorphism*

$$H_k(H_j(O)) \cong H_j(H_k(O)) \quad (3.8)$$

fixing O and the named stable unitaries v_j, v_k .

Proof. Consider $C = H_k(H_j(O))$. Corollary 3.3 places both $H_j(O)$ and $H_k(O)$ inside C . Compose the expectations $C \rightarrow H_j(O) \rightarrow O$, and call the result E_O . Its restriction to each smaller adjunction is that adjunction's expectation onto O , as can be checked on its reduced words.

We prove that the two smaller adjunctions are free over O : an alternating product of elements from their respective kernels of E_O has zero E_O -expectation. Approximate each centered $H_k(O)$ -element by finite sums of nonempty reduced k -words over O . Such approximations follow by reducing a polynomial approximation and subtracting its O -expectation. Keep the centered $H_j(O)$ -elements as coefficients.

When two k -words are joined across a centered element $b \in H_j(O)$, the new internal coefficient has the form aba' , where $a, a' \in O$. Its expectation onto O is $aE_O(b)a' = 0$, so its expectation onto either D_i^k is also zero. The resulting word is therefore a nonempty reduced k -word over $H_j(O)$, including at every new opposite-sign turn. Initial and final $H_j(O)$ -coefficients are allowed. Its expectation onto $H_j(O)$ vanishes. Passing to norm limits proves the freeness assertion. An alternating product with no k -block is a single centered j -element and is already killed by E_O .

The individual expectations and this freeness determine all joint O -valued moments. Indeed, center the factors of a joint word; the fully alternating centered term vanishes, and every other term has fewer blocks after its O -coefficients are absorbed into neighbors. The resulting recursion

is symmetric in j and k . Applying the same argument in the reverse construction gives equal scalar trace moments for every polynomial in O, v_j, v_k . Faithfulness and (3.6) give the desired isomorphism. $\square$

By Lemma 2.8, restrictions and compositions of these expectations agree whenever they have the same range and preserve the specified trace.

Lemma 3.5. *Let λ be a nonzero limit ordinal, and let $(O_\beta)_{\beta < \lambda}$ be an increasing family of unital C^* -algebras with compatible faithful tracial states. Suppose that fixed subalgebras D_1, D_2 lie in every O_β , with a fixed trace-preserving isomorphism $\theta : D_1 \rightarrow D_2$ and compatible trace-preserving conditional expectations $E_i^\beta : O_\beta \rightarrow D_i$. Use the embeddings of Corollary 3.3 to identify the reduced adjunctions $H(O_\beta)$ with the same stable unitary v . Then, with*

$$O = \overline{\bigcup_{\beta < \lambda} O_\beta}, \quad R = \overline{\bigcup_{\beta < \lambda} H(O_\beta)},$$

the expectations extend to $O \rightarrow D_i$, and $R = H(O)$ with its specified generators and trace.

Proof. The compatible norm-one expectations extend continuously to O and retain complete positivity, bimodularity and trace preservation. Denote these extensions by E_i . The compatible traces extend faithfully to both completions. For example, the quotient of R by its trace-null ideal is injective, hence isometric, on every earlier faithfully tracial algebra. It is therefore isometric on their dense union and on R , so that ideal is zero. The same argument applies to O .

We have $R = C^*(O, v)$, and the unitary relation persists. Consider a nonempty reduced word with coefficients in O . Approximate its finitely many coefficients in one common earlier O_β . At a turn requiring $E_i(a) = 0$, replace an approximation b by $b - E_i^\beta(b)$. This remains in O_β , is exactly centered, and differs from a by at most twice the original error. Thus the given word is a norm limit of reduced words at earlier stages, all with trace zero. Proposition 3.1 identifies R with $H(O)$. This finite-coefficient argument requires no assumption on the cofinality of λ . $\square$

4 Joining two classes while preserving unique extensions

We now choose the abelian subalgebras in the reduced HNN construction. The resulting inclusion has two features needed in the iteration: every old pure state extends uniquely, and precisely the prescribed two old representation classes merge. The proof must control arbitrary old pure states, even though the defining data consist of only two peaking sequences.

Theorem 4.1 (The bridge). *Let A be a unital C^* -algebra with a faithful tracial state τ and no finite-dimensional irreducible representations. For $i = 1, 2$, let $(p_n^i)_{n \geq 0}$ be a decreasing sequence of projections, with $p_0^i = 1$, peaking to a pure state f_i . Suppose that $f_1 \not\sim f_2$ and*

$$\tau(p_n^1) = \tau(p_n^2) = t_n, \quad 1 = t_0 > t_1 > \cdots > 0, \quad t_n \longrightarrow 0.$$

Put $D_i = C^(p_n^i : n \geq 0)$ and let $\theta : D_1 \rightarrow D_2$ be determined by $\theta(p_n^1) = p_n^2$. Form the reduced HNN extension $H(A)$ of Proposition 3.1 using these data and their trace-preserving conditional expectations. Then every pure state h of A has a unique state extension $\tilde{h}$ to $H(A)$. These extensions are pure, and for all pure states h, k of A ,*

$$\tilde{h} \sim \tilde{k} \iff h \sim k \text{ or } \{[h], [k]\} = \{[f_1], [f_2]\}.$$

The extension has a faithful tracial state extending τ and a faithful trace-preserving conditional expectation onto A . If A is simple, then $H(A)$ is simple.

Proof. Lemma 2.9 supplies the expectations onto D_i . The differences $p_n^i - p_{n+1}^i$ are nonzero, mutually orthogonal projections, so each D_i is the algebra of continuous functions on $\mathbb{N}_0 \cup \{\infty\}$. Matching these differences defines θ , and the equality of trace values makes it trace preserving. Thus the reduced HNN construction applies. Its trace and expectation already give the corresponding assertions in the theorem. The work is to prove the assertions about pure states.

Pass to the two-by-two construction. Use the notation of Lemma 3.2 and Proposition 3.1:

$$F = M_2(A), \quad G = M_2(D_1), \quad D = D_1 \oplus D_2, \quad Z = F *_D G.$$

Write $E_D^F : F \rightarrow D$ for the expectation, $e_i = e_{ii}$ for the common diagonal projections, and $s = e_{21}^G$. Recall that $H(A) = e_1 Z e_1$, that $Z = M_2(H(A))$ using the matrix units of F , and that $v = e_{12}^F s$ satisfies $vp_n^1 v^* = p_n^2$ in the corner. Put

$$P_n^i = p_n^i e_i \in F, \quad \phi_i(a) = f_i(a_{ii}) \quad (a \in F).$$

Here $p_n^i e_i$ denotes the matrix with entry p_n^i in position (i, i) and zero elsewhere. The projections P_n^i peak to ϕ_i : their zeroth term is e_i , so a state taking value one on them is supported on that corner and restricts there to f_i . The states ϕ_1, ϕ_2 are pure and inequivalent, by matrix amplification of the GNS representations of f_1, f_2 . Also F has no finite-dimensional irreducible representations. Indeed, any such representation would restrict along $a \mapsto \text{diag}(a, a)$ to a nonzero finite-dimensional representation of A , whose image would supply a finite-dimensional irreducible representation of A .

We shall prove unique extension and the exclusion of unintended joins for $F \subset Z$, and then return to the corner. By Lemma 2.2, it suffices to use ϕ_i as the representative of its distinguished equivalence class. More explicitly, conjugating by a unitary of F bijects the sets of state extensions. We first establish uniqueness for these representatives and for every pure state outside their two classes; only afterward will we use the resulting unique lifts.

Consider pure states h, k of F satisfying one of the two tests

$$h = k, \quad \text{or} \quad h \not\sim k \quad \text{and} \quad \{[h], [k]\} \neq \{[\phi_1], [\phi_2]\}. \quad (4.1)$$

Whenever either state belongs to $[\phi_i]$, take it to be ϕ_i itself. Choose decreasing positive contractive excision nets $x = x_\lambda$ for h and $y = y_\nu$ for k as in Lemma 2.4; at ϕ_i , choose the sequence P_n^i by Lemma 2.5. All limits involving both x and y below are on the product of the two index sets.

We will show that $x_\lambda W y_\nu \rightarrow 0$ in norm for every nonempty word W in (3.7). When $h = k$, this will force all state extensions to agree; for the inequivalent pairs in (4.1), it will keep the lifted representations inequivalent. The stable unitary will then join the designated pair.

A norm estimate for reduced words. We claim that

$$\|x a_0 b_1 a_1 \cdots b_m a_m y\| \longrightarrow 0 \quad (4.2)$$

for every $m \geq 1$, every $b_j \in \{s, s^*\}$, and every $a_j \in F$ with

$$E_D^F(a_j) = 0 \quad (0 < j < m). \quad (4.3)$$

By the spanning description (3.7), these words, together with F , span a dense subspace of Z .

Let P_G be the orthogonal projection of the reduced free-product Hilbert D -module K onto the summands starting with a factor in L_G° . For $d \in F$, the decomposition $K = L_F \otimes_D K^{(F)}$ gives

$$\|d P_G\| \leq \|E_D^F(d^* d)\|^{1/2}. \quad (4.4)$$

Indeed, a vector $\eta \in P_G K$ has the form $1 \otimes \eta$ in this decomposition, and

$$\|d \otimes \eta\|^2 = \|\langle \eta, E_D^F(d^* d) \eta \rangle\| \leq \|E_D^F(d^* d)\| \|\eta\|^2.$$

The expectation norm here is the norm in D .

For $b \in \{s, s^*\}$ write $b = e_{\ell(b)} b e_{r(b)}$. Thus $(\ell(s), r(s)) = (2, 1)$ and $(\ell(s^*), r(s^*)) = (1, 2)$. For a word in (4.2), put

$$\ell = \ell(b_1), \quad r = r(b_m), \quad w = b_1 a_1 \cdots a_{m-1} b_m.$$

Each letter of this centered alternating core creates a new leg when applied, from right to left, to a word not starting with G . Therefore

$$(1 - P_G)w(1 - P_G) = 0, \quad w = P_G w + (1 - P_G)w P_G.$$

Set $d = x a_0 e_\ell$ and $d' = e_r a_m y$. We obtain

$$\|d w d'\| \leq \|d P_G\| \|w\| \|d'\| + \|d\| \|w\| \|P_G d'\|. \quad (4.5)$$

Using (4.4) and its adjoint version, it follows that (4.2) holds whenever both

$$\begin{aligned} \|E_D^F(e_\ell a_0^* x^2 a_0 e_\ell)\| &\longrightarrow 0, \\ \|E_D^F(e_r a_m y^2 a_m^* e_r)\| &\longrightarrow 0. \end{aligned} \quad (4.6)$$

This sufficient criterion does not require the pair of endpoint states to satisfy (4.1).

When an endpoint decays. In the first line of (4.6), increase x^2 to x . The resulting elements are a decreasing net of positive continuous functions on the ℓ th convergent sequence in the spectrum of D ; they vanish on the other component. At a finite atom $Q = P_n^\ell - P_{n+1}^\ell$, their pointwise limit is

$$\frac{\tau_F(Q a_0^* r_h a_0 Q)}{\tau_F(Q)} = 0.$$

Here r_h is the support projection in F^{**} and τ_F is the normalized matrix trace, extended normally to F^{**} . The numerator vanishes by traciality and Lemma 2.7; for example

$$0 \leq \tau_F(r_h a_0 Q a_0^*) \leq \|a_0 Q a_0^*\| \tau_F(r_h) = 0.$$

At the point ∞ , the limit is $\phi_\ell(a_0^* r_h a_0)$. It is zero if $h \not\sim \phi_\ell$, because the normal extension of the ϕ_ℓ representation annihilates r_h , as in Lemma 2.6. If $h = \phi_\ell$, the support identity instead gives

$$\phi_\ell(a_0^* r_h a_0) = |\phi_\ell(a_0)|^2.$$

Thus the first endpoint decays unless $h = \phi_\ell$, and it still decays in that case when $\phi_\ell(a_0) = 0$.

The uniform convergence used here is Dini's theorem for monotone nets [9, Theorem 1.5, p. 306]. We recall its compactness argument. For a fixed $\varepsilon > 0$, the open sets where a chosen function is below ε cover the compact spectrum. Choose a finite subcover, and then an index dominating its finitely many indices. Monotonicity makes every later function uniformly smaller than ε . The second line of (4.6) is symmetric: it decays unless $k = \phi_r$, and also decays in that case when $\phi_r(a_m) = 0$.

Remove the matched endpoints. Call the left endpoint matched when $h = \phi_\ell$, and the right endpoint matched when $k = \phi_r$. At each matched endpoint write

$$a_0 = \phi_\ell(a_0)1 + a_0^0, \quad \text{or} \quad a_m = \phi_r(a_m)1 + a_m^0.$$

Expand the word using these decompositions. In each resulting term, an unmatched endpoint already satisfies its estimate in (4.6), and a matched endpoint with its zero-value coefficient does as well. We treat the scalar terms by moving the corresponding peak through its adjacent letter:

$$P_n^{\ell(b)}b = bP_n^{r(b)}, \quad bP_n^{r(b)} = P_n^{\ell(b)}b. \quad (4.7)$$

These identities hold in G : the diagonal embedding of D_2 uses θ^{-1} , so the two projections have the same matrix entry p_n^1 there.

In the following displays, an untransferred endpoint coefficient means the coefficient chosen in the expansion. For $m \geq 2$, after suppressing its scalar coefficient, a left scalar term becomes

$$P_n^{\ell(b_1)}b_1a_1 \cdots b_ma_my = b_1(P_n^{r(b_1)}a_1b_2 \cdots b_ma_my).$$

The peak now belongs to the other distinguished state, and b_1 is a bounded outside factor. A right scalar choice similarly moves $P_t^{r(b_m)}$ to $P_t^{\ell(b_m)}$ and leaves b_m as a bounded factor on the right. When both ends are transferred and $m \geq 3$, the resulting expression, again up to its scalar coefficient, is

$$b_1(P_n^{r(b_1)}a_1b_2 \cdots b_{m-1}a_{m-1}P_t^{\ell(b_m)})b_m.$$

Suppose at least one core letter remains after the transfers. Each newly exposed endpoint coefficient is one of the original internal a_j . By (4.3), $\phi_1(a_j) = \phi_2(a_j) = 0$: these states evaluate the two components of $E_D^F(a_j)$ at ∞ . Thus every new endpoint satisfies its estimate, whether or not its transferred state matches the adjacent letter. Endpoints that were not transferred satisfy their estimates by the preceding expansion. Applying (4.6) to the remaining word therefore proves decay of each such term. This covers no transfers, one transfer when $m \geq 2$, and two transfers when $m \geq 3$, including a remaining core of one letter. The sufficient endpoint criterion requires no assumption that the transferred pair of states satisfies (4.1).

It remains to check the cases in which the core disappears. If $m = 1$, both ends cannot match: the range and source of s or s^* are different, so double matching would be the designated pair excluded in (4.1). For a single left transfer, say $h = \phi_i$, the remaining compression, up to a scalar and a bounded outside letter, is

$$P_n^j a_1 y_\nu, \quad j \neq i.$$

The state k is inequivalent to ϕ_j . In the equal-state test it equals ϕ_i ; in the other test, equivalence to ϕ_j would again be the excluded designated pair. Hence this expression tends to zero by Lemma 2.6. A single right transfer is symmetric.

Finally let $m = 2$ and transfer at both ends. Both original endpoints matched, so (4.1) forces $h = k = \phi_i$ for some i . Since each letter switches the two matrix indices,

$$\ell(b_1) = r(b_2) = i, \quad r(b_1) = \ell(b_2) = j \neq i.$$

The remaining compression is $P_n^j a_1 P_t^j$, with $\phi_j(a_1) = 0$. If $q = \min(n, t)$, nestedness and excision give

$$\|P_n^j a_1 P_t^j\| \leq \|P_q^j a_1 P_q^j\| \rightarrow 0.$$

This is convergence on the product of the two sequence indices. Together with cross-excision, whose conclusion also holds on the product net, these cases prove (4.2) in full.

Unique extensions and the exclusion of other joins. In the equal-state test, let ψ be any state of Z extending h . Its GNS vector is fixed by every x_λ and y_ν : a positive contraction x of state value one satisfies $\|(1 - \pi_\psi(x))\xi_\psi\|^2 \leq \psi(1 - x) = 0$. Consequently, for every tested word W ,

$$\psi(W) = \psi(x_\lambda W y_\nu) \rightarrow 0.$$

Its values on F are already prescribed. The dense span above therefore proves uniqueness; existence also follows directly by composing h with the expectation onto F . Conjugacy by unitaries in F now establishes uniqueness for every old pure state, including all states in the two distinguished classes. Their unique extensions are pure by Lemma 2.1.

Next take an inequivalent pair allowed in (4.1). If its unique lifts were equivalent, realize them by unit vectors ξ_h, ξ_k in one irreducible representation π of Z . The represented excision elements fix the corresponding vectors, so

$$\langle \pi(z)\xi_k, \xi_h \rangle = \langle \pi(x_\lambda z y_\nu)\xi_k, \xi_h \rangle.$$

For $z \in F$, the right side tends to zero by cross-excision; for a tested nonempty word it tends to zero by (4.2). Thus the matrix coefficient vanishes on all of Z . This contradicts cyclicity of ξ_k in the irreducible representation. Unitary conjugacy within the old classes extends this exclusion to all representatives. Equivalent old states, on the other hand, have equivalent unique lifts by Lemmas 2.2 and 2.1.

Return to the corner and join the designated pair. States of $A = e_1 F e_1$ identify with states of F taking value one on e_1 , by composition with compression. The same correspondence holds between $H(A) = e_1 Z e_1$ and Z , and it commutes with restriction. Every extension to Z of an e_1 -supported state of F is still e_1 -supported. Hence this correspondence bijects extension sets and transfers unique extension to $A \subset H(A)$.

Under $F = M_2(A)$ and $Z = M_2(H(A))$, these supported states have the matrix amplifications of their corner GNS representations, with the representing vector in the first summand. Purity and equivalence of classes are therefore preserved. The matrix-flip unitary of F moves the representative ϕ_2 from the second summand to the first, so its class corresponds to $[f_2]$, just as the class of ϕ_1 corresponds to $[f_1]$. The exclusion of all other joins thus descends to the corner.

To see that the designated join actually occurs, use the stable unitary relation. The state $\tilde{f}_2 \circ \text{Ad } v$ takes value one on every p_n^1 , so its restriction to A is f_1 . Unique extension, now proved, gives

$$\tilde{f}_1 = \tilde{f}_2 \circ \text{Ad } v.$$

This proves the asserted equivalence criterion. If A is simple, Lemma 2.10 applies to the faithful expectation and the unique-extension property, proving that $H(A)$ is simple. $\square$

5 Closed subiterations and countable determination

We now iterate the bridge. The main point is that a pure state on the resulting algebra is uniquely determined on a suitable separable subalgebra. To prove this, we first show that one may retain any collection of bridge steps containing the dependencies of each of its members, and then perform all omitted steps afterward.

5.1 States and traces along a chain

A chain $(B_\alpha)_{\alpha \leq \lambda}$ of unital inclusions is *continuous* if $B_\gamma = \overline{\bigcup_{\alpha < \gamma} B_\alpha}$ at every nonzero limit ordinal $\gamma \leq \lambda$.

Lemma 5.1. *Let $(B_\alpha)_{\alpha \leq \lambda}$ be a continuous chain whose successor inclusions have property (U). Then every inclusion $B_\alpha \subset B_\beta$, for $\alpha \leq \beta \leq \lambda$, has (U). The unique lifts of equivalent pure states are equivalent.*

Proof. At a successor, unique extensions compose, and the intermediate lift is pure by Theorem 2.1. At a limit, two extensions agree on each earlier algebra, hence on their dense union. Existence follows from state extension. Transfinite induction gives the first assertion. For the second, a unitary implementing equivalence in the original algebra also conjugates the unique lifts; use Theorem 2.2. $\square$

Lemma 5.2. *Let $(B_\alpha)_{\alpha \leq \lambda}$ be continuous, with λ a nonzero limit ordinal. Suppose pure states h, k of B_λ restrict to inequivalent pure states of B_α for all α in some tail below λ . Then $h \not\sim k$.*

Proof. Otherwise realize h, k by unit vectors ξ, η in one irreducible representation π of B_λ . For each earlier α in the given tail, the cyclic subspaces

$$M_\alpha = \overline{\pi(B_\alpha)\xi}, \quad N_\alpha = \overline{\pi(B_\alpha)\eta}$$

are reducing and carry inequivalent irreducible representations of B_α . The orthogonal projection onto N_α , restricted to M_α , is an intertwiner, so it is zero. Thus $\langle \pi(a)\xi, \eta \rangle = 0$ for every $a \in B_\alpha$. Continuity gives the same equality for every $a \in B_\lambda$, contradicting cyclicity of ξ . $\square$

We will also use two elementary permanence facts. A continuous union of simple unital algebras is simple: a proper quotient is injective, and hence isometric, on every earlier algebra, so it is isometric on their dense union and its completion. Compatible faithful tracial states remain faithful at the limit. Indeed, the extended trace's null ideal has zero intersection with every earlier algebra; the quotient by this ideal is again isometric on a dense union, so the ideal is zero. At successors, simplicity follows from Theorem 2.10 whenever (U) and a faithful conditional expectation are available.

5.2 The initial algebra and its subiterations

Let

$$A_0 = \bigotimes_{m=1}^{\infty} M_2, \quad p_n = e_{11}^{\otimes n} \otimes 1 \quad (n \geq 1), \quad p_0 = 1.$$

Here A_0 is the norm closure of its unital matrix subalgebras M_{2^n} , with the compatible normalized matrix traces.

Lemma 5.3. *The algebra A_0 is separable, simple and infinite-dimensional, its trace is faithful, and (p_n) peaks to a pure state f_0 . Moreover, $\tau(p_n) = 2^{-n}$. Every unital C^* -algebra containing A_0 has no finite-dimensional irreducible representations.*

Proof. Simplicity and faithfulness follow from the permanence facts above applied to the matrix stages. Separability and infinite dimension are immediate from those stages. A state taking value one on p_n must restrict to the vector state of the first basis vector on M_{2^n} . These vector states are compatible, so they determine exactly one state on the dense union and its completion. This proves peaking and purity. The trace formula is the normalized matrix trace formula.

A nonzero finite-dimensional irreducible representation of an algebra containing A_0 restricts to a unital representation of A_0 . Its kernel is zero by simplicity, which would embed the infinite-dimensional algebra A_0 in a finite-dimensional algebra. $\square$

Consider a continuous chain $(X_\beta)_{\beta \leq \delta}$ starting from $X_0 = A_0$. At an *active* index $j < \delta$, perform the bridge of Theorem 4.1 using (p_n) and a decreasing sequence of projections (q_n^j) that peaks in X_j to a state inequivalent to the p_n -state, with

$$q_0^j = 1, \quad \tau(q_n^j) = 2^{-n}.$$

Write v_j for its stable unitary, so $X_{j+1} = C^*(X_j, v_j)$. At every other index set $X_{j+1} = X_j$. The traces are the compatible traces supplied by the bridges. By Theorem 5.1, (p_n) peaks at every stage. For each active j , (q_n^j) peaks at every stage from j onward, and its state is equivalent to the p_n -state from stage $j+1$ onward.

For $T \subset \delta$, define the unital subalgebra

$$A[T] = C^*(A_0, \{v_j : j \in T, j \text{ active}\}) \subset X_\delta.$$

Thus $A[\beta] = X_\beta$ when the ordinal β is regarded as the set of smaller ordinals. Every element of X_δ lies in $A[T]$ for some countable T : approximate it by a sequence of polynomials, each involving finitely many stable unitaries.

Choose countable sets $I_j \subset j$ such that $q_n^j \in A[I_j]$ for every n at each active j , and set $I_j = \emptyset$ otherwise. Call T *dependency-closed* if

$$j \in T \implies I_j \subset T. \tag{5.1}$$

Every countable set is contained in a countable dependency-closed set: repeatedly adjoin the dependencies of its members and take the union over the natural numbers.

Lemma 5.4. *For every dependency-closed $T \subset \delta$, the algebras $A[T \cap \beta]$, $\beta \leq \delta$, form a continuous bridge iteration in the inherited order, with identity steps at indices outside T . In particular, $A[T]$ is simple and has a faithful tracial state. If T is countable, $A[T]$ is separable.*

Proof. Let $k \in T$ be active. Both sequences for its bridge lie in $A[T \cap k]$, by (5.1). The original expectations onto their generated abelian algebras restrict to this smaller base, since their ranges already lie in it. The restriction property Theorem 3.3 identifies the algebra obtained by adjoining v_k with that reduced HNN extension.

The sequences peak to pure states of the smaller base by Theorem 2.3, applied to its inclusion in X_k . That same lemma gives each of these particular states its unique extension to X_k . If the two smaller states were equivalent, Theorem 2.2 and uniqueness would make their extensions equivalent, contrary to the choice of the original active step. The smaller base contains A_0 , so Theorem 5.3 excludes finite-dimensional irreducibles. All hypotheses of Theorem 4.1 therefore hold in the inherited iteration.

Continuity follows because a polynomial uses only finitely many indices. Simplicity follows by induction from Theorem 2.10 and the limit observation above. Faithfulness follows either from the restricted trace on X_δ or from the same permanence argument. Finally, a countable set of stable unitaries together with the separable algebra A_0 generates a separable algebra. $\square$

5.3 Moving omitted steps to the end

The preceding lemma proves unique extension along the inherited iteration. We need the stronger assertion that it holds from the entire inherited algebra into the original result.

Theorem 5.5. *For every dependency-closed $S \subset \delta$, the inclusion $A[S] \subset X_\delta$ has property (U).*

Proof. We will insert the omitted indices, in their original order, after $A[S]$. Recall that the ordinal j is the set of indices below j , whereas $j + 1$ also contains the index j itself. Fix an active $j \notin S$, and for $j + 1 \leq \beta \leq \delta$ set

$$O_\beta = A[j \cup (S \cap \beta)], \quad R_\beta = A[(j + 1) \cup (S \cap \beta)]. \quad (5.2)$$

These index sets are dependency-closed. Initially, $O_{j+1} = X_j$ and $R_{j+1} = X_{j+1}$; afterward both chains perform precisely the inherited steps with indices in S .

The omitted pair stays inequivalent on O_β . Both (p_n) and (q_n^j) peak on every O_β by Theorem 2.3, since they peak in X_δ . Their states are initially inequivalent. Suppose $k > j$ is an active member of S , and they are still inequivalent on O_k . The q_n^k -sequence also lies in O_k and peaks there. Each of these three particular states has a unique extension to X_k , by Theorem 2.3; this does not use the theorem being proved.

In X_k , the q_n^j -state is equivalent to the p_n -state, because step j has already been performed. The q_n^k -state is inequivalent to that state, because step k is active. Equivalence on O_k would pass to these particular unique extensions. Consequently, on O_k the q_n^k -state is equivalent to neither the p_n -state nor the q_n^j -state. The k -bridge joins only the classes of its designated pair, by Theorem 4.1. It therefore does not join the two states of the omitted j -bridge. At limits their inequivalence persists by Theorem 5.2. Transfinite induction proves the claim.

The omitted adjunction commutes past the inherited steps. Write H_j for the reduced HNN adjunction with the fixed data $(p_n), (q_n^j)$. We claim, respecting the trace and named generators,

$$R_\beta \cong H_j(O_\beta) \quad (j + 1 \leq \beta \leq \delta). \quad (5.3)$$

This is true initially. At an active $k \in S$ above j , the data for both adjunctions lie in O_k . The expectation from $R_k = H_j(O_k)$ onto O_k , followed by either k -data expectation, is a trace-preserving expectation onto that data algebra. By Theorem 2.8, it agrees with the expectation supplied there by the corresponding peaking sequence. The same observation applies with j and k interchanged. Thus the reordering isomorphism Theorem 3.4 applies and gives

$$H_k(H_j(O_k)) \cong H_j(H_k(O_k)),$$

proving the next instance of (5.3). Other successor steps are identities. Figure 1 displays the two orders at an active index $k \in S$.

$$\begin{array}{ccc} O_k & \xrightarrow{\text{adjoin } v_j} & H_j(O_k) = R_k \\ \text{adjoin } v_k \downarrow & & \downarrow \text{adjoin } v_k \\ H_k(O_k) = O_{k+1} & \xrightarrow{\text{adjoin } v_j} & R_{k+1} \cong H_k(H_j(O_k)) \\ & & \cong H_j(H_k(O_k)) \end{array}$$

Figure 1: Reordering two adjunctions whose data all lie in O_k . The arrows are inclusions. Both paths give the same faithfully tracial algebra, under an isomorphism fixing O_k, v_j, v_k .

At a limit, the isomorphisms already constructed agree on overlaps because they fix the named generators. The expectations from the O_β onto the two fixed abelian algebras generated by (p_n)

and (q_n^j) also agree on overlaps, by Theorem 2.8. Applying Theorem 3.5 to these fixed data identifies the closure of the preceding $H_j(O_\beta)$ with H_j of the limit base, proving (5.3) there.

At $\beta = \delta$ we have $O_\delta = A[S \cup j]$ and $R_\delta = A[S \cup (j+1)]$. The two sequences still peak to inequivalent states on O_δ , by the first part of the proof. Thus Theorem 4.1 applies to (5.3), giving (U) for

$$A[S \cup j] \subset A[S \cup (j+1)].$$

If $j \in S$ or j is inactive, this inclusion is the identity. The continuous chain $(A[S \cup j])_{j \leq \delta}$ starts at $A[S]$ and ends at X_δ ; continuity follows because a polynomial involves only finitely many indices outside S . Now Theorem 5.1 proves the theorem. $\square$

5.4 A separable algebra determines each pure state

The separable pure-restriction argument below is closely related to [1, Proposition 2.3 and concluding Remark 1]. Here we arrange that the separable algebra is dependency-closed; Theorem 5.5 then adds uniqueness of the full extension.

Lemma 5.6. *For every pure state h of X_δ , there is a countable dependency-closed $S \subset \delta$ such that $h|_{A[S]}$ is pure and h is its unique state extension to X_δ .*

Proof. Construct increasing countable dependency-closed sets S_n , starting with $S_0 = \emptyset$. Choose a countable dense subset D_n of $A[S_n]$. For each $a \in D_n$ and each integer $m \geq 1$, Theorem 2.4 supplies a positive contraction $x_{a,m}$ in X_δ satisfying

$$h(x_{a,m}) = 1, \quad \|x_{a,m}(a - h(a)1)x_{a,m}\| < 1/m.$$

Each witness has countable support. Adjoin all these supports to S_n and close under dependencies to obtain S_{n+1} .

Set $S = \bigcup_n S_n$ and $B = A[S]$. Suppose $h|_B = th_1 + (1-t)h_2$ for states h_1, h_2 and $0 < t < 1$. Each h_i takes value one on every witness $x_{a,m}$. A positive contraction of state value one fixes the state's GNS vector, so $h_i(a) = h_i(x_{a,m}ax_{a,m})$ and $h_i(x_{a,m}^2) = 1$. Hence

$$|h_i(a) - h(a)| = |h_i(x_{a,m}(a - h(a)1)x_{a,m})| < 1/m.$$

Letting m tend to infinity shows that both states agree with h on every D_n . Their union is dense in B , so $h|_B$ is pure. Its unique extension is h by Theorem 5.5. $\square$

Corollary 5.7. *Put $\mathfrak{c} = 2^{\aleph_0}$. If $|\delta| \leq \mathfrak{c}$, then X_δ has at most $\mathfrak{c}$ pure states.*

Proof. There are at most

$$\max(\aleph_0, |\delta|)^{\aleph_0} \leq \mathfrak{c}^{\aleph_0} = \mathfrak{c}$$

countable subsets of δ . For each one, the corresponding algebra $A[S]$ is separable and has at most $\mathfrak{c}$ states: values on a countable dense subset determine a state. By Theorem 5.6, every pure state of X_δ is the unique extension of one such restriction. There are at most $\mathfrak{c} \cdot \mathfrak{c} = \mathfrak{c}$ possibilities. $\square$

6 The construction in ZFC

We now choose the bridge steps so that every pure state eventually joins the class of the state peaked by (p_n) . Countable determination ensures that every pure state of the final algebra will have appeared early enough to be treated.

Theorem 6.1 (Kishimoto–Ozawa–Sakai). *Let B be a separable C^* -algebra. If pure states h, k of B have GNS representations with the same kernel, there is an automorphism γ of B such that $h \circ \gamma = k$. In particular, this holds for every pair of pure states of a separable simple C^* -algebra.*

This is [8, Theorem 1.1], which in fact provides an asymptotically inner automorphism. We apply it only to the separable algebras $A[S]$ of Theorem 5.4.

Proof of Theorem 1.1. Set $\kappa = \mathfrak{c}^+$. Every ordinal below κ has cardinality at most $\mathfrak{c}$, and every countable subset of κ is bounded: otherwise κ would be a countable union of sets of cardinality at most $\mathfrak{c}$, a contradiction.

Fix the schedule first. Before constructing any algebras, choose an injection

$$\sigma : \kappa \times \kappa \longrightarrow \kappa, \quad \sigma(\beta, \eta) \geq \beta. \quad (6.1)$$

To do so, enumerate $\kappa \times \kappa$ in order type κ . When a pair (β, η) is encountered, choose an unused ordinal at least β . Fewer than κ ordinals have been used, whereas the tail above β has cardinality κ . This defines the entire schedule without referring to any states.

Construct the algebras in ordinal order. Build a continuous bridge iteration $(X_\beta)_{\beta \leq \kappa}$ starting from the algebra of Theorem 5.3. For each $\beta < \kappa$, once X_β has been constructed, choose a list

$$(h_{\beta, \eta})_{\eta < \kappa}$$

containing every pure state of X_β , with repetitions allowed. This is possible by Theorem 5.7, since $|\beta| \leq \mathfrak{c}$ for $\beta < \kappa$.

At stage $j < \kappa$, let f be the state of X_j peaked by (p_n) . If j is outside the range of σ , make the identity step. Otherwise write $j = \sigma(\beta, \eta)$ and let g be the unique lift of $h_{\beta, \eta}$ to X_j . The list for β is already available because $\beta \leq j$; when $\beta = j$, it is chosen before the successor step. If $g \sim f$, again make the identity step.

Suppose $g \not\sim f$. By Theorem 5.6, choose a countable dependency-closed $S \subset j$ such that $g|_{A[S]}$ is pure and uniquely determines g on X_j . The restriction $f|_{A[S]}$ is pure by Theorem 2.3. The algebra $A[S]$ is separable, simple and unital by Theorem 5.4. Applying Theorem 6.1, choose an automorphism γ of $A[S]$ with

$$(g|_{A[S]}) \circ \gamma = f|_{A[S]}, \quad q_n^j = \gamma(p_n) \quad (n \geq 0).$$

If a state on $A[S]$ takes value one on every q_n^j , composing it with γ gives the unique state taking value one on every p_n . Thus (q_n^j) peaks to $g|_{A[S]}$. By Theorem 5.5, it peaks to g on X_j as well.

The trace requirement deserves a separate check. For each n , γ carries the unital matrix stage $M_{2^n} \subset A_0$ onto a unital matrix subalgebra of $A[S]$, and carries its minimal projection p_n to q_n^j . The ambient trace restricts to the unique normalized trace on this matrix algebra. Consequently,

$$\tau(q_n^j) = 2^{-n} = \tau(p_n).$$

This argument does not require an automorphism to preserve the specified trace on all of $A[S]$.

We can therefore perform the bridge of Theorem 4.1 with $(p_n), (q_n^j)$, and set $I_j = S$. At an identity step set $I_j = \emptyset$. At limit stages take the norm completion of the union. Each successor satisfies (U) and has a faithful trace-preserving expectation onto its base. The results of Section 5 therefore apply throughout the recursion.

All choices are set-sized. At every stage the algebra is generated by at most κ elements, so its density is at most κ and its underlying cardinality at most $\lambda = \kappa^{\aleph_0}$. Fix a set of cardinality λ . Algebra structures on its subsets, together with their norms, traces, embeddings, states, automorphisms and lists, form a set of codes. Well-order that set to make the required choices; recode successor algebras with their embeddings and use norm inductive limits. Ordinary choice and transfinite recursion in ZFC suffice.

Every final pure state is treated. Let $A = X_\kappa$ and let h be any pure state of A . By Theorem 5.6, there is a countable dependency-closed $S \subset \kappa$ such that $h|_{A[S]}$ is pure and has unique extension h to A . Boundedness of S gives $\beta < \kappa$ with $A[S] \subset X_\beta$. The state $h|_{X_\beta}$ is the unique extension of $h|_{A[S]}$ to X_β : any other extension could itself be extended to A , contradicting uniqueness there. Hence $h|_{X_\beta}$ is pure.

Choose η with $h_{\beta,\eta} = h|_{X_\beta}$ and put $j = \sigma(\beta, \eta)$. Its lift to X_j is $h|_{X_j}$. At stage j this lift either already belongs to the p_n -class or is joined to it by the bridge. By Theorem 5.1, the equivalence persists to A . Thus all pure states of A have unitarily equivalent GNS representations.

Every nonzero irreducible representation π of the unital algebra A is unital: the range of $\pi(1)$ is a nonzero closed invariant subspace. Every nonzero vector is cyclic by irreducibility, so each unit vector defines a pure state whose GNS representation is equivalent to π . The p_n -state supplies one such representation. Therefore A has exactly one nonzero irreducible representation up to unitary equivalence.

Finally, A is simple and has a faithful tracial state by Theorem 5.4 and the limit permanence facts. It is unital and contains the infinite-dimensional algebra A_0 . An algebra $K(H)$ is unital only when H is finite-dimensional: the identity on an infinite-dimensional Hilbert space is not compact, as an infinite orthonormal sequence has no norm-convergent subsequence. The finite-dimensional case, including $H = 0$, cannot be isomorphic to A . This proves the theorem. $\square$

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